



CYBERSECURITY ROADMAP

Cybersecurity Engineering

A comprehensive, end-to-end roadmap covering fundamental IT skills, OS administration, networking, CTF platforms, certifications, cryptography, web security, DFIR, cloud security, and automation.

DURATION

24 weeks

LEVEL

Intermediate

TOPICS

36

SECTIONS

10

Cybersecurity Engineering is a vital discipline dedicated to protecting digital systems, networks, applications, and sensitive data from cyber threats, unauthorized access, and malicious attacks through defensive operations, ethical hacking, and cloud security control frameworks.

Full interactive roadmap & more free learning paths at www.roadmaplibrary.com

WHAT'S INSIDE

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Every section below includes hands-on subtopics with day-by-day pacing, plus hand-picked courses, docs, and practice platforms.

HOW TO USE THIS ROADMAP

A Structured Path, Not a Checklist

This roadmap breaks Cybersecurity Engineering into 10 progressive sections, moving from IT fundamentals and networking up through offensive security, DFIR, and cloud automation. Work through them in order for the strongest foundation, or jump to a section you need most. Each topic is split into short subtopics with a suggested day range, so pair the reading and videos with real practice — spin up a VM, run the labs, and use the linked CTF platforms to reinforce what you learn.

Days 1–2

Suggested pacing

 Course Article Project Docs Video

Tip: treat the estimated 24 weeks as a floor, not a ceiling — slow down on sections like networking and web security, since they underpin almost everything that follows.

01. Fundamental IT Skills & Hardware

2 topics · 4 lessons

01 Hardware Components, Connections & Troubleshooting

Understanding hardware architecture, connection interfaces, and OS-independent troubleshooting is essential for diagnosing compromised endpoints and physically securing hardware.

Days 1–2

Hardware Components & Connection Types

Study CPU architectures, RAM, motherboard buses, storage controllers (NVMe/SATA), and connectivity interfaces including USB, Ethernet, NFC, Bluetooth, and Infrared.

Days 3–4

OS-Independent Troubleshooting & Productivity Suites

Learn hardware diagnostics, POST boot error codes, BIOS/UEFI configuration, and security considerations across productivity suites (MS Office, Google Workspace, iCloud).

RESOURCES



[CompTIA A+ Hardware Training - Professor Messer](#)



[Hardware Troubleshooting Guide - ComputerHope](#)



02 CTF Platforms & Security Certifications

Hands-on CTF (Capture the Flag) platforms provide practical exploit training, while industry certifications validate foundational and advanced security knowledge.

Days 1–2

Hands-on CTFs (HackTheBox, TryHackMe, picoCTF)

Practice hands-on penetration testing and privilege escalation on TryHackMe, HackTheBox, VulnHub, picoCTF, and SANS Holiday Hack Challenges.

Days 3–4

Beginner & Advanced Security Certifications

Explore career certification paths from entry-level (CompTIA A+, Network+, Security+, Linux+, CCNA) to advanced credentials (CEH, OSCP, CISSP, CISA, CISM, GIAC, CREST).

RESOURCES



[TryHackMe - Complete Beginner Path](#)



[HackTheBox Academy Labs](#)



[picoCTF Practice Gym](#)





SANS Holiday Hack Challenge Archive



02. Operating Systems & Administration

1 topic · 2 lessons

01 Windows, Linux & macOS System Administration

Operating system internals dictate access controls, file permissions, background processes, and audit log tracking across Windows, Linux, and macOS platforms.

Days 1–2

OS Installation, Navigation & User Permissions

Master GUI and CLI navigation, installation procedures, local account creation, file permissions (ACLs vs POSIX), and software package management across Windows, Linux, and macOS.

Days 3–4

System CRUD Operations & OS Troubleshooting

Perform file CRUD operations via terminal, inspect system services (systemctl, services.msc), execute diagnostic commands, and audit system event logs.

RESOURCES

[Linux Command Line Basics for Security - Linux.org](#)[Windows Sysinternals Utilities Documentation](#)

03. Computer Networking & Protocols

2 topics · 5 lessons

01 OSI Model, Topologies & Network Services

Networking protocols govern digital communication. Master OSI layers, packet routing, IP addressing, subnetting, network topologies, and infrastructure services.

Days 1–2

OSI Model, TCP/IP Handshakes & Subnetting

Study OSI 7 layers, TCP 3-way handshakes, IP addressing (IPv4/IPv6, CIDR notation, subnet masks, loopback, default gateways), and private vs public IP ranges.

Days 3–4

Network Topologies & Infrastructure Services

Compare Star, Mesh, Ring, and Bus network topologies. Understand core network infrastructure services: DHCP, DNS, NTP, IPAM, switches, routers, and VPN gateways.

Days 5–6

Network Protocols (SSH, RDP, HTTPS, SFTP, DNS)

Master secure vs unsecure protocols: SSH vs Telnet, SFTP/SCP vs FTP, HTTPS vs HTTP, SSL/TLS handshakes, DNSSEC, LDAPS, SRTP, and S/MIME.

RESOURCES

► Video

[NetworkChuck - Network Fundamentals Course](#)

≡ Docs

[RFC 793 - Transmission Control Protocol](#)

02

Virtualization & Network Diagnostics Tools

Virtualization enables isolated testing environments, while network diagnostic tools allow security analysts to inspect live packet traffic and open ports.

Days 1–2

Virtualization Technologies (VMware, Proxmox, ESXi)

Configure Type-1 (ESXi, Proxmox) and Type-2 (VirtualBox, VMware Workstation) hypervisors, manage HostOS vs GuestOS, and deploy isolated testing VMs.

Days 3–4

Network & Packet Analysis Tools (Nmap, Wireshark, tcpdump)

Master network diagnostic and analysis utilities: ping, traceroute, nslookup, dig, netstat, ipconfig, arp, route, nmap port scanning, tcpdump, and Wireshark PCAPs.

RESOURCES

≡ Docs

[Proxmox VE Official Documentation](#)

≡ Docs

[Nmap Network Scanning Book & Command Reference](#)

≡ Docs

[Wireshark User's Guide](#)

04. Authentication, Security Skills & Architecture

2 topics · 4 lessons

01

Authentication Methods, SSO & Directory Services

Identity management prevents unauthorized entry. Study single sign-on (SSO), Kerberos, LDAP, RADIUS, X.509 certificates, and multi-factor authentication (MFA).

Days 1–2

Directory Services & SSO (Kerberos, LDAP, RADIUS)

Understand enterprise authentication protocols: Active Directory LDAP/LDAPS, Kerberos tickets (TGT), RADIUS AAA servers, SAML/OAuth SSO, and local auth hashes.

Days 3–4

Multi-Factor Authentication (MFA) & PKI Tokens

Implement 2FA/MFA token verification (TOTP/HOTP), hardware security keys (FIDO2/WebAuthn), digital certificates, and privilege elevation policies.

RESOURCES

≡ Docs

[Active Directory & Kerberos Protocol Guide](#)

Course

[Auth0 - Identity & Access Management Fundamentals](#)

02

Security Principles, Zero Trust & Defense in Depth

Core security architecture balances risk, accessibility, and defense layers. Learn the CIA Triad, Defense in Depth, Zero Trust architecture, and risk management.

Days 1–2

CIA Triad, Risk Management & Defense in Depth

Master Confidentiality, Integrity, Availability (CIA Triad), risk assessment definitions, business continuity, backups, and multi-layered Defense in Depth.

Days 3–4

Zero Trust Architecture, DMZ & Network Segmentation

Implement Zero Trust (never trust, always verify), micro-segmentation, perimeter DMZs, air-gapped isolation, honeypots, and Role-Based Access Control (RBAC).

RESOURCES

≡ Docs

[CISA Zero Trust Maturity Model Version 2.0](#)

≡ Docs

[NIST Special Publication 800-53 Security Controls](#)

05. Web Application Security & OWASP Top 10

1 topic · 2 lessons

01

OWASP Top 10, Web Exploitation & Malware

Web applications and malware payloads represent the primary initial access vectors. Study OWASP vulnerabilities, malware types, and reverse engineering principles.

Days 1–2

OWASP Top 10 Web Vulnerabilities (SQLi, XSS, CSRF)

Audit and exploit web vulnerabilities: SQL Injection (SQLi), Cross-Site Scripting (XSS), CSRF, Directory Traversal, Buffer Overflows, and Broken Access Control.

Days 3–4

Malware Analysis & Reverse Engineering Fundamentals

Understand malware types (Ransomware, Trojans, Keyloggers, Rootkits, Worms, Spyware), static code analysis, obfuscation techniques, and reverse engineering basics.

RESOURCES



Course

[PortSwigger Web Security Academy](#)

Docs

[OWASP Top 10 Web Application Vulnerabilities](#)

06. Threat Intelligence, Frameworks & Standards

1 topic · 2 lessons

01

MITRE ATT&CK, SIEM, SOAR & OSINT

Threat intelligence frameworks standardize attack categorization. Master MITRE ATT&CK, Cyber Kill Chain, SIEM/SOAR log aggregation, and OSINT gathering.

Days 1–2

Cyber Security Frameworks (MITRE ATT&CK, Diamond Model)

Study threat frameworks: MITRE ATT&CK matrix (Tactics, Techniques, Procedures), Diamond Model of Intrusion Analysis, and Lockheed Martin Cyber Kill Chain.

Days 3–4

Compliance Standards (NIST, ISO 27001, CIS) & SIEM/SOAR

Understand security compliance (NIST CSF/RMF, ISO 27001, CIS Controls), SIEM log ingestion (Splunk/Elastic), SOAR automated playbooks, and OSINT recon.

RESOURCES



Docs

[MITRE ATT&CK Matrix for Enterprise](#)

Docs

[Splunk Security Essentials Reference Guide](#)

07. Digital Forensics & Incident Response (DFIR)

1 topic · 2 lessons

01 Incident Response Lifecycle & Forensic Discovery Tools

When security breaches occur, DFIR teams contain threats and analyze digital evidence. Learn the IR lifecycle, forensic imaging, and forensic analysis tools.

Days 1–2

Incident Response Process (NIST / SANS Framework)

Execute the 6 IR phases: Preparation, Identification, Containment, Eradication, Recovery, and Lessons Learned. Manage evidence chain of custody.

Days 3–4

Forensic Analysis Tools (FTK Imager, Autopsy, Volatility)

Analyze RAM dumps and disk images using FTK Imager, Autopsy, WinHex, memdump, Volatility 3, and parse event logs, Syslog, and NetFlow captures.

RESOURCES

≡ Docs [SANS DFIR Incident Response Cheat Sheet](#)



≡ Docs [Volatility 3 Memory Forensics Documentation](#)



08. System Hardening, Defense & Endpoint Security

1 topic · 2 lessons

01 OS Hardening, Endpoint Security & EDR

Hardening reduces attack surfaces on endpoints and servers. Study endpoint protection (EDR/DLP), firewalls, Group Policy, patch management, and port blocking.

Days 1–2

Operating System & Network Hardening

Implement OS hardening, Group Policy Objects (GPO), Network Access Control (NAC), MAC filtering, port blocking, DNS sinkholes, and patch management.

Days 3–4

Endpoint Security (EDR, DLP, NGFW, Sandboxing)

Deploy EDR (Endpoint Detection & Response), Data Loss Prevention (DLP), Next-Gen Firewalls (NGFW), Host IPS (HIPS), and malware sandboxing (VirusTotal, Joe Sandbox, Any.run).

RESOURCES

≡ Docs [CIS Benchmarks OS Hardening Standards](#)



⚙️
Project

[VirusTotal Malware Inspection Engine](#)



[Joe Sandbox Automated Analysis Platform](#)

09. Cyber Attacks, Vectors & Threat Actors

1 topic · 2 lessons

01

Social Engineering, Network Attacks & Password Exploits

Understanding threat actor techniques enables proactive defense. Study social engineering vectors, network attacks, DoS/DDoS, and password cracking strategies.

Days 1–2

Social Engineering & Phishing Vectors

Identify social engineering tactics: Phishing, Vishing, Whaling, Smishing, Spam, Shoulder Surfing, Tailgating, Impersonation, Watering Hole, and Typo Squatting.

Days 3–4

Network & Web Attacks (DDoS, MITM, Spoofing, Pass-the-Hash)

Analyze attacks: DoS/DDoS, Man-in-the-Middle (MITM), ARP/IP Spoofing, Evil Twin, DNS Poisoning, VLAN Hopping, Deauth, Replay, and Pass-the-Hash.

RESOURCES

**Article**[Social Engineering Framework & Attack Vectors](#)**Docs**[MITRE ATT&CK Technique T1566: Phishing](#)

10. Cloud Security & Security Scripting

1 topic · 2 lessons

01

Cloud Security, IaC & Automation Scripting

Modern environments span hybrid cloud platforms and require security automation scripting in Python, Bash, PowerShell, and Go.

Days 1–2

Cloud Security Models (AWS, Azure, GCP, IaC, Serverless)

Understand cloud security shared responsibility model, service models (IaaS, PaaS, SaaS), cloud deployment (Public, Private, Hybrid), Infrastructure as Code (IaC), AWS/Azure/GCP, and S3/cloud storage.

Days 3–4

Security Automation Languages (Python, Bash, PowerShell, Go)

Write security automation scripts using Python, Bash, PowerShell, Go, and C++. Utilize LOLBAS (Living Off The Land), GTFOBins, and WADComs for security auditing.

RESOURCES

≡ Docs

[AWS Security Reference Architecture \(SRA\)](#)

Project

[GTFOBins - Unix Binaries Exploitation Index](#)

Project

[LOLBAS - Living Off The Land Binaries](#)

Project

[WADComs - Windows AD Command Search Engine](#)



Keep Going

You've made it through the full Cybersecurity Engineering roadmap. Consistency beats intensity — work through each section at your own pace, build real labs, and revisit the resources as you go.

CONTINUE LEARNING

<https://roadmaplibrary.com/#roadmaps>

Browse more free, structured roadmaps at www.roadmaplibrary.com